ByteBoost Workshop: Project Description

Shape-Constrained Boosted Symbolic Regression for Interpretable Neural Scaling Laws: A Cross-Testbed Investigation

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1 Field of science

Computer Science: Machine Learning and Artificial Intelligence, specifically the foundations of large language models (neural scaling laws) and interpretable machine learning via symbolic regression. Secondary field: High-Performance Computing and computer architecture (cross-platform performance characterization).

2 Abstract / summary

Training a modern AI language model can cost millions of dollars, so researchers use scaling laws to predict performance from model size and training data before committing resources. These equations guide major hardware and budget decisions, but their mathematical form is usually chosen manually and then fitted to measured results.

Our project uses symbolic regression to discover the scaling-law equation directly from data. Symbolic regression searches many candidate formulas and returns an interpretable equation rather than a black-box model. Unconstrained searches, however, often produce formulas that fit existing measurements but extrapolate unrealistically, predicting negative error or worse performance with more data.

We therefore require every candidate formula to satisfy admissibility axioms: larger models and additional data cannot increase error; improvements must diminish with scale; error must remain positive and approach an irreducible floor; excess loss must decay as a power law; and the formula must remain smoothly well-defined down to the smallest scales of interest. We mathematically certify these properties across the full continuum domain of interest, not only at measured points (Sections 4.2, 4.4 and 4.6).

The formula is built incrementally from a standard scaling law, adding one correction at a time while preserving all guarantees (Propositions 1 and 2 and Theorem 1). We also prove that the corrections improve fit without changing the irreducible error floor or the asymptotic rate of improvement (Theorem 1). The primary search path *penalizes* axiom violations inside a genetic-programming fitness and drives them toward zero while bounding any remaining shortfall (Section 4.7 and Proposition 3); a second path *rejects* inadmissible corrections outright and is a workshop extension of the soft engine (Sections 4.4 and 4.5).

The project has two computational bottlenecks. Model training will use Cerebras CS-3 systems on PSC's Neocortex platform and will be compared with completed NVIDIA GPU runs (Sections 3.1 and 3.2). Formula discovery is a large CPU-bound search dominated by interval-arithmetic certification. Because it is highly parallel and relies mainly on scalar or imperfectly vectorized Python (with optional JAX acceleration), Stony Brook's high-core-count AMA27 AmpereOne cluster is well suited to the workload (Sections 3.1 and 4.4).

Deliverables will include a certified scaling-law formula fitted to our training data, an open-source discovery library, and a concise methods report (Section 5.2). An initial set of trained models is already public (Section 3.5).

3 Key requirements

3.1 Testbed hardware (requested)

Code skeleton: `src/modeling/training/`, `src/systems/hpc/`, `src/search/`

- **Neocortex (PSC):** Cerebras CS-3 wafer-scale systems for the model-pretraining grid in Sections 3.4 and 3.5 (`https://www.psc.edu/resources/neocortex/`). Each CS-3 is built around a Wafer Scale Engine 3 (WSE-3) with on-chip SRAM and extreme memory bandwidth, so a model in our N range can stay on a single wafer without multi-GPU model parallelism. We use Neocortex only for pretraining; loss curves are compared to our existing NVIDIA GPU runs (Section 3.2).
- **AMA27 (Stony Brook Supercomputing Center / ACCESS):** Arm-based AmpereOne A192-32M cluster for the CPU-bound symbolic-regression search of Section 4.3 and the certification inner loop of Section 4.4. Per the ACCESS resource description, phase 1 deploys 312 single-socket compute nodes, each with a 192-core AmpereOne A192-32M CPU and 384 GB DDR5, linked by 200 Gbps NDR InfiniBand, with a 5 PB GPFS filesystem (`https://support.access-ci.org/affinity-groups/ama27`). AMA27 is the Stony Brook Arm testbed featured in ByteBoost 2.0 (in place of the retired Ookami A64FX system). We do *not* rely on A64FX-style 512-bit SVE or HBM: the GP/IA workload is population-parallel and largely scalar Python DualInterval evaluation, so the useful AMA27 properties are Arm ISA diversity versus our AMD EPYC (x86) baselines, high core counts per node, and dense node count for concurrent candidate evaluation.

3.2 Baseline hardware (already available to us)

Code skeleton: `src/systems/hpc/profiling.py`, `src/modeling/training/trainer.py`

GPU baselines (pretraining comparisons): 200 NVIDIA GH200 superchips on DeltaAI at NCSA, plus a local four-GH200 cluster. **CPU baselines (search/certification comparisons):** three 256-core AMD EPYC (x86) machines, against which we will profile the AMA27 AmpereOne port.

3.3 Software and tools

Code skeleton: `src/modeling/training/`, `src/search/`, `src/constraints/certificates/`

PyTorch 2.x for pretraining (distributed data-parallel training; optional mixed precision), with `uv` for environment management and Weights & Biases for experiment logging [1]. Symbolic search uses `gplearn` [2] with a custom fitness that adds soft interval-arithmetic axiom penalties (Section 4.7); an optional PySR backend provides an unconstrained alternate engine for ablation. Certificates are evaluated with a pure-Python forward-mode *DualInterval* interval-arithmetic layer (Section 4.4); JAX with custom JVPs is an optional acceleration path, not required. A hard reject filter that discards inadmissible candidates during search (Section 4.4) is a workshop extension built on the same certificate primitives.

3.4 Models

Code skeleton: `src/modeling/models/config.py`, `src/modeling/models/transformer.py`

Decoder-only transformer language models (a compact post-LN causal stack suitable for scaling grids), spanning roughly 10M–1.4B parameters, across several tokens-per-parameter ratios ($M = D/N$). Workshop ports may adopt LLaMA-style blocks [3], FlashAttention, and μP / μTransfer [4]; the core method only requires consistent (N, D, L) measurements in the Step Law [5] / overtraining [6] regimes.

3.5 Datasets

Code skeleton: `src/scaling/data/scaling_dataset.py`, `src/scaling/data/corpora.py`

Pretraining corpora used to generate new grid points include Wikipedia [7], RedPajama [8], and related open text mixtures (optional FineWeb-Edu [9] for workshop extensions). An already-public `colinear_scaling_models` collection on Hugging Face [10] supplies trained checkpoints and loss curves from an initial DeltaAI grid for the symbolic-regression track.

4 Method summary (technical)

Code skeleton: `src/scaling/`, `src/constraints/`, `src/search/`; map: Appendix B

Symbols used below are collected in Appendix A; the student skeleton layout is summarized in Appendix B.

4.1 Setup

Code skeleton: `src/scaling/setup/configuration.py`, `src/scaling/setup/constants.py`

Let N be the parameter count, D the training-token count, and $\mathcal{H} = \{h_1, \ldots, h_m\}$ a finite set of remaining hyperparameters (in the concrete 4D grids used here, typically width-multiplier / weight-decay style axes and learning rate); see also Appendix A. For each coordinate $h \in \{N, D\} \cup \mathcal{H}$, let H_h be its finite set of admissible values and $\phi_h\colon H_h \to \mathbb{R}_{>0}$ a strictly monotone preprocessing map; write $\phi_i := \phi_{h_i}$ for brevity. In the search implementation, genetic-programming features are the logarithms $\log \phi_h(h)$ (so stage corrections are fit in log-coordinates and mapped back to raw-scale derivatives by the chain rule; Section 4.4). The configuration space is the product $\mathbb{H} := \prod_{h \in \{N,D\} \cup \mathcal{H}} H_h$, with typical element $\mathbf{h} \in \mathbb{H}$ (so N, D, h_i denote the corresponding coordinates of $\mathbf{h}$). Let $L\colon \mathbb{H} \to \mathbb{R}$ be validation loss; we seek $\hat{L} \approx L$. We are given a labeled dataset $\mathcal{D} = \{(\mathbf{h}_\ell, L(\mathbf{h}_\ell))\}_{\ell=1}^{n} \subset \mathbb{H} \times \mathbb{R}$. For each scale variable $x \in \{N, D\}$ set $x_{\min} := \min H_x > 0$ and $x_{\max} := \max H_x$, treat x as continuous on $[x_{\min}, \infty)$, and take all derivatives with the other coordinates held fixed. Write $\widetilde{\mathbb{H}}$ for the corresponding continuum domain: each scale ranges over $[x_{\min}, \infty)$ while non-scale coordinates still range over their finite H_h, so $\mathbb{H} \subset \widetilde{\mathbb{H}}$. The labeled loss L is observed only on $\mathbb{H}$; the approximant $\hat{L}$ is evaluated on $\widetilde{\mathbb{H}}$ when stating shape constraints (Section 4.2).

4.2 Admissibility axioms

Code skeleton: `src/constraints/axioms/admissibility.py`

$\hat{L}$ is *admissible* ($\hat{L} \in \mathcal{S}$) if it satisfies (A1)–(A6) below. Where an axiom mentions a scale variable x, it is imposed for each $x \in \{N, D\}$ with the other coordinates held fixed. Although the labeled grid $\mathbb{H}$ is finite, $\hat{L}$ is constrained as a real function on the continuum domain $\widetilde{\mathbb{H}}$ of Section 4.1, so that the derivatives in (A1)–(A2) and the limits in (A3)–(A4) are well-defined.

(A1) **Monotone decrease.** $\dfrac{\partial \hat{L}}{\partial x} \le 0$ for all $x \ge x_{\min}$.

(A2) **Diminishing returns.** $\dfrac{\partial^2 \hat{L}}{\partial x^2} \ge 0$ for all $x \ge x_{\min}$. (Together with (A1), this is the usual diminishing-returns shape: further increases in x yield weakly smaller loss reductions.)

(A3) **Irreducible loss.** There is a positive *joint floor* L_∞ and, along each axis, a *marginal floor* L_x^∞:
(i) *Joint floor.* There exists $L_\infty > 0$ such that $\hat{L}(\mathbf{h}) > L_\infty$ for every $\mathbf{h} \in \widetilde{\mathbb{H}}$.

(ii) *Marginal floor.* For each $x \in \{N, D\}$ and every fixed choice of the other coordinates, the single-axis limit

$$L_x^\infty \;:=\; \lim_{t\to\infty} \hat{L}(\mathbf{h})\big|_{x=t} \tag{1}$$

exists and is finite, and satisfies $L_x^\infty \geq L_\infty$. (The value L_x^∞ depends on the frozen coordinates; the notation suppresses that dependence.)

In the stage-0 Chinchilla law, L_∞ equals the joint limit as all scales $\to \infty$ (namely E), while a single-axis marginal is in general strictly larger.

(A4) **Asymptotic power-law decay.** Relative to the marginal floor (1),

$$\lim_{t\to\infty} \frac{\log\big(\hat{L}(\mathbf{h})\big|_{x=t} - L_x^\infty\big)}{\log t} \;=\; c_x \qquad \text{for some } c_x < 0. \tag{2}$$

The excess $\hat{L}(\mathbf{h})|_{x=t} - L_x^\infty$ is eventually positive (so the logarithm is defined) whenever the limit in (A3)(ii) is approached from above, which follows from (A1) when the floor is not attained at finite t.

(A5) **Positivity.** $\hat{L}(\mathbf{h}) > 0$ for every $\mathbf{h} \in \widetilde{\mathbb{H}}$. Since $L_\infty > 0$, (A5) is implied by the joint-floor inequality in (A3); we retain it as an explicit physical prior (the soft path folds it into the floor score v_{irred}).

(A6) **Graceful saturation.** $\hat{L}$ is twice continuously differentiable (C^2) in x on $[x_{\min}, \infty)$ —so that the second derivative in (A2) exists and is continuous—and $\hat{L}(x_{\min}, \cdot)$ is finite. (Expression-tree corrections used later are in fact C^∞ on $(0, \infty)$. The inequality $\hat{L}(x_{\min}, \cdot) > L_\infty$ is already required by (A3)(i).)

These encode physical priors on loss surfaces that unconstrained symbolic regression routinely violates. In particular, (A4) must use L_x^∞ rather than L_∞: along a single axis the excess over the joint floor need not vanish, so a power-law rate would not be identifiable against L_∞.

4.3 Boosted construction

Code skeleton: `src/search/baselines/`, `src/search/residuals/`, `src/search/expression/`

The ensemble is built stagewise, $\hat{L}^{(j)} = \hat{L}^{(j-1)} + \hat{L}_j$, targeting an *admissibility invariant*: $\hat{L}^{(j)} \in \mathcal{S}$ for every $j = 0, \ldots, K$ (Section 4.2), enforced exactly by the hard filter and recovered by the soft path whenever $V = 0$ (Section 4.7 and Theorem 1). Throughout, $\hat{L}_j$ (subscript) is the stage-j correction and $\hat{L}^{(j)}$ (superscript) is the cumulative ensemble through stage j, so in particular $\hat{L}^{(0)} = \hat{L}_0$. Stage 0 is a generalized Chinchilla law, fit by nonlinear least squares with positive parameter constraints. Concrete instances include a 2D law in (N, D) and a 4D law that adds two hyperparameter axes (e.g. weight decay and learning rate). The 4D hyperparameter terms may be power laws of the same form as in (3), or—matching U-shaped loss in lr/wd—quadratic penalties in $\log \phi_{\text{lr}}(\text{lr})$ and $\log \phi_{\text{wd}}(\text{wd})$ (shape axioms still act only on N and D):

$$\hat{L}^{(0)} \;=\; \hat{L}_0 \;=\; \frac{A}{N^\alpha} + \frac{B}{D^\beta} + \sum_{i=1}^{m} \frac{C_i}{\phi_i(h_i)^{\gamma_i}} + E, \qquad A, B, C_i, E, \alpha, \beta, \gamma_i > 0. \tag{3}$$

Proposition 1 (Stage-0 admissibility)**.** *The law* (3) *with all coefficients and exponents strictly positive lies in $\mathcal{S}$, with joint floor $L_\infty = E$ and asymptotic exponents $c_N^{(0)} = -\alpha$, $c_D^{(0)} = -\beta$ (the rates c_x in (A4) of Section 4.2 along N and D).*

(See Appendix C.5 for the proof.) Each later stage $j \geq 1$ fits Huber pseudo-residuals of the current ensemble, with clipping threshold

$$\delta_j \;:=\; \text{median}_{\ell=1,\ldots,n} \big| L(\mathbf{h}_\ell) - \hat{L}^{(j-1)}(\mathbf{h}_\ell) \big| \tag{4}$$

so that outlier runs cannot dominate:

$$\tilde{r}_j^{(\ell)} \;:=\; \begin{cases} L(\mathbf{h}_\ell) - \hat{L}^{(j-1)}(\mathbf{h}_\ell) & \text{if } \big|L(\mathbf{h}_\ell) - \hat{L}^{(j-1)}(\mathbf{h}_\ell)\big| \le \delta_j, \\ \delta_j \cdot \operatorname{sign}\big(L(\mathbf{h}_\ell) - \hat{L}^{(j-1)}(\mathbf{h}_\ell)\big) & \text{otherwise.} \end{cases} \tag{5}$$

Corrections are sought by genetic-programming search over expression trees with primary operators $\{+, -, \times, \div\} \cup \{\mathrm{pow}_p : p \in \mathcal{P}\}$ for a finite set $\mathcal{P} \subset \mathbb{R} \setminus \{0\}$ (where $\mathrm{pow}_p(z) = |z|^p$ for $z > 0$) and bounded depth; optional unary helpers such as inv/log/sqrt may be enabled for ablations. Trees are evaluated on *log-features* $\log \phi_h(h)$; raw-scale derivatives needed by (A1)–(A2) are recovered by the chain rule (Section 4.4).

4.4 Stagewise admissibility

Code skeleton: `src/constraints/axioms/stage_conditions.py`, `src/constraints/certificates/`, `src/search/hard_path/`

Checking that the full ensemble $\hat{L}^{(j)}$ obeys (A1)–(A6) of Section 4.2 is a continuum statement on $\widetilde{\mathbb{H}}$ and along each scale half-line. The next claim reduces that check to conditions on the increment alone.

$$\begin{gathered} \text{(i) } \frac{\partial \hat{L}_j}{\partial x} \le -\frac{\partial \hat{L}^{(j-1)}}{\partial x} \quad \text{for all } x \ge x_{\min}, \\ \text{(ii) } \frac{\partial^2 \hat{L}_j}{\partial x^2} \ge -\frac{\partial^2 \hat{L}^{(j-1)}}{\partial x^2} \quad \text{for all } x \ge x_{\min}, \\ \text{(iii) } \hat{L}_j(\mathbf{h}) > -\hat{L}^{(j-1)}(\mathbf{h}) \quad \text{for all } \mathbf{h} \in \widetilde{\mathbb{H}}, \\ \text{(iv) } \hat{L}^{(j-1)}(\mathbf{h}) + \hat{L}_j(\mathbf{h}) > L_\infty \quad \text{for all } \mathbf{h} \in \widetilde{\mathbb{H}}, \\ \text{(v) } \hat{L}_j(\mathbf{h})\big|_{x=t} = O\big(t^{c_{x,j}}\big) \text{ as } t \to \infty \text{ for some } c_{x,j} < c_x^{(0)}, \\ \text{(vi) } \hat{L}_j \in C^\infty \text{ on } [x_{\min}, x_{\max}] \text{ with } \big|\hat{L}_j(x_{\min}, \cdot)\big| < \infty. \end{gathered} \tag{6}$$

When $L_\infty > 0$, (iv) already implies (iii); both are retained so positivity of the ensemble and preservation of the joint floor remain explicit.

Proposition 2 (Stagewise equivalence)**.** *Given $\hat{L}^{(j-1)} \in \mathcal{S}$ with joint floor L_∞, the update $\hat{L}^{(j)} = \hat{L}^{(j-1)} + \hat{L}_j$ lies in $\mathcal{S}$ with the same joint floor if and only if $\hat{L}_j$ satisfies* (6)*(i)–(vi) for each $x \in \{N, D\}$.*

(See Appendix C.1 for the proof.) Conditions (i)–(iv) and (vi) are still quantified over continua ((i)–(ii) on the half-line $x \ge x_{\min}$; (iii)–(iv) on $\widetilde{\mathbb{H}}$), and (v) is an asymptotic statement: none of these can be decided by sampling finitely many points of $\mathbb{H}$. They become machine-checkable because every GP proposal g is a concrete finite expression tree built from $\{+, -, \times, \div\} \cup \{\mathrm{pow}_p : p \in \mathcal{P}\}$. Write $F := \hat{L}^{(j-1)} + g$. Two complementary certificates decide whether g preserves (6): interval checks establish (i)–(iv) and (vi) on the compact box $\mathcal{I}_x$ covering the observed grid, while the structural check decides (v) exactly (and thereby controls the $x \to \infty$ tail that the box does not cover).

(a) Sound interval-arithmetic checks (conditions (i)–(iv), (vi) on a compact slice of $\widetilde{\mathbb{H}}$). Over the compact box $\mathcal{I}_x := [x_{\min}, x_{\max}]$ (other coordinates ranging over their finite H_h), the product of boxes is a compact continuum subset of $\widetilde{\mathbb{H}}$ covering the observed scale range; the half-line $x > x_{\max}$ is handled by the structural test (b) below. Forward-mode automatic differentiation composed with interval arithmetic [11, 12] evaluates g bottom-up on interval operands and returns guaranteed enclosures $\underline{f} \le f \le \overline{f}$ on $\mathcal{I}_x$ for $f \in \{F, \partial F/\partial x, \partial^2 F/\partial x^2\}$, in time linear in the tree size. (The second-derivative enclosure is what makes (A2) / stage (ii) checkable; (A6) asks only for C^2, while the operator set of

Section 4.3 in fact yields C^∞ on $(0,\infty)$.) When g is represented on log-features, DualInterval evaluation is performed in log-coordinates and first/second derivatives are mapped to the raw axis x by the chain rule before applying (7). A candidate is accepted when, for each $x \in \{N, D\}$,

$$\overline{\partial F/\partial x} \le 0, \quad \underline{\partial^2 F/\partial x^2} \ge 0, \quad \underline{F} > L_\infty, \quad \overline{F} < \infty. \tag{7}$$

Because the enclosures are sound, (7) is a *sufficient* certificate that the corresponding continuum conditions hold *everywhere* on $\mathcal{I}_x$, not merely at sampled points of $\mathbb{H}$ (Lemma 1 and Appendix C.2): $\overline{\partial F/\partial x} \le 0$ forces $\partial F/\partial x \le 0$ on $\mathcal{I}_x$ (hence (i) on the box), and likewise for (ii); $\underline{F} > L_\infty > 0$ forces (iv) on the continuum configurations covered by the product of boxes, and hence also (iii) / positivity (A5); $\overline{F} < \infty$ forces the finiteness half of (vi), while C^2 (in fact C^∞) smoothness of g on $\mathcal{I}_x$ follows from the operator set and discharges (A6) on the box. The test is conservative: a failure of (7) does *not* prove that g fails (6) (interval over-approximation can reject a truly admissible update), but every accepted g is guaranteed to preserve admissibility on $\mathcal{I}_x$. That one-sided decidability is exactly what a hard filter needs, and evaluating (7) on each GP candidate is the inner-loop hot spot, the direct target of the AMA27 performance work. In the primary soft path (Section 4.7), the same enclosures are converted to continuous violation scores rather than used as a reject gate.

(b) Exact structural checks (condition (v), and prerequisites for it). Define the asymptotic exponent $\operatorname{ord}(g, x)$ recursively on the tree: constants and leaves other than x contribute 0; $\mathrm{pow}_p(x)$ contributes p; products add exponents; quotients subtract; sums take the maximum. Then (v) holds with $c_{x,j} = \operatorname{ord}(g, x)$ precisely when $\operatorname{ord}(g, x) < c_x^{(0)}$ for each $x \in \{N, D\}$ (Lemma 2 and Appendix C.2). We additionally require that each scale variable $x \in \{N, D\}$ appears as a leaf of g (otherwise the exponent in x is undefined or vacuously 0). Both tests are exact, finite walks over a finite tree, with no floating-point search over $\widetilde{\mathbb{H}}$.

4.5 Algorithm

Code skeleton: `src/search/boosting/`

Putting the pieces together, Algorithm 1 builds the ensemble of Section 4.3. The *primary* soft path of Section 4.7 selects each correction by minimizing the penalized fitness (10); the workshop hard-filter extension instead constrains the stage argmin by certificates (a)–(b) of Section 4.4 (written explicitly in the pseudocode below).

Algorithm 1 Shape-constrained boosted symbolic regression

Require: Dataset $\mathcal{D} = \{(\mathbf{h}_\ell, L(\mathbf{h}_\ell))\}_{\ell=1}^n$, stages K
Ensure: Ensemble $\hat{L}^{(K)}$ (admissible under the hard filter; soft path aims for $V = 0$ at every stage)

Fit $\hat{L}_0$ via (3) on $\mathcal{D}$; set $\hat{L}^{(0)} \leftarrow \hat{L}_0$
for $j = 1, \ldots, K$ **do**
  Compute δ_j via (4)
  Compute $\tilde{r}_j^{(\ell)}$ for all $\ell = 1, \ldots, n$ via (5)
  Find $\hat{L}_j$ by soft search (10) **or** by $\arg\min_g \frac{1}{n} \sum_{\ell=1}^n \big(\tilde{r}_j^{(\ell)} - g(\mathbf{h}_\ell)\big)^2$ subject to certificates (a)–(b) for (6) ▷ soft primary; hard filter is the workshop extension
  $\hat{L}^{(j)} \leftarrow \hat{L}^{(j-1)} + \hat{L}_j$
end for
return $\hat{L}^{(K)}$

4.6 Guarantee

Code skeleton: `src/constraints/guarantee/invariants.py`

The hard path of Algorithm 1 preserves admissibility, the joint floor, and the stage-0 rates:

Theorem 1 (Hard-path boosting guarantee)**.** *Assume $\hat{L}_0$ is the admissible stage-0 law of Proposition 1 (so $L_\infty = E$). If every later correction $\hat{L}_j$ $(j \geq 1)$ is accepted by certificates (a)–(b) of Section 4.4, then for all j: $\hat{L}^{(j)} \in \mathcal{S}$; writing $L_\infty^{(j)}$ for the joint floor of $\hat{L}^{(j)}$, one has $L_\infty^{(j)} = E$; and the asymptotic exponents are preserved,*

$$\lim_{t\to\infty} \frac{\log\big(\hat{L}^{(j)}(\mathbf{h})\big|_{x=t} - L_x^\infty\big)}{\log t} \;=\; c_x^{(0)}, \tag{8}$$

where L_x^∞ is the marginal floor of $\hat{L}^{(j)}$ along x (equal to that of $\hat{L}_0$).

Boosting therefore improves fit *without* corrupting the interpretable constants researchers actually quote (proof in Appendix C.3).

4.7 Soft-constrained variant

Code skeleton: `src/search/soft_path/violations.py`, `src/search/soft_path/fitness.py`, `src/search/soft_path/backends.py`

The primary implemented path uses per-axiom violation scores on the candidate ensemble $F := \hat{L}^{(j-1)} + g$, folded into the `gplearn` fitness (Section 3.3), rather than a hard reject gate. Using the interval enclosures of Section 4.4(a) and the structural quantities of (b), define the soft scores of Appendix A:

$$\begin{aligned}
v_{\text{mono}}(g) &= \max_{x\in\{N,D\}} \max\Big(0,\ \overline{\partial F/\partial x}\Big), && \text{(monotonicity gap)},\\
v_{\text{conv}}(g) &= \max_{x\in\{N,D\}} \max\Big(0,\ -\underline{\partial^2 F/\partial x^2}\Big), && \text{(convexity gap)},\\
v_{\text{irred}}(g) &= \max\big(0,\ L_\infty - \underline{F}\big), && \text{(floor / positivity gap)},\\
v_{\text{decay}}(g) &= \max_{x\in\{N,D\}} \max\big(0,\ \operatorname{ord}(g,x) - c_x^{(0)} + \varepsilon_{\text{decay}}\big), && \text{(power-law gap)},\\
v_{\text{leaf}}(g) &= \big|\{x \in \{N,D\} : x \notin \operatorname{leaves}(g)\}\big|, && \text{(missing scale leaves)}.
\end{aligned} \tag{9}$$

Here $\varepsilon_{\text{decay}} > 0$ is a fixed margin so that $v_{\text{decay}}(g) = 0$ forces the strict inequality $\operatorname{ord}(g,x) \leq c_x^{(0)} - \varepsilon_{\text{decay}} < c_x^{(0)}$ required by (6)(v); $\operatorname{leaves}(g)$ is the set of variable leaves of the expression tree g. Every score is nonnegative, and $v_{\text{mono}} = v_{\text{conv}} = v_{\text{irred}} = v_{\text{leaf}} = 0$ iff the corresponding hard certificate passes (for v_{decay}, zero is sufficient but slightly stronger than (v) because of the margin). Writing a for an axiom index ranging over $\{\text{mono}, \text{conv}, \text{irred}, \text{decay}, \text{leaf}\}$ and $\lambda_a > 0$ for its penalty weight, the aggregate violation is $V(g) = \sum_a v_a(g)$. Stage j then minimizes the penalized fitness

$$F_j(g) \;=\; \underbrace{\frac{1}{n}\sum_{\ell=1}^{n}\big(\tilde{r}_j^{(\ell)} - g(\mathbf{h}_\ell)\big)^2}_{\text{data fit}} \;+\; \underbrace{\sum_a \lambda_a\, v_a(g)^2}_{\text{axiom penalty}}, \tag{10}$$

where F_j is a scalar fitness (distinct from the ensemble map F above), the first term is mean squared error of g against the Huber pseudo-residuals (5), and the second term penalizes axiom gaps (squared so the penalty gradient vanishes at zero violation). Candidates that already satisfy (6) therefore incur no penalty pressure.

Proposition 3 (Soft-path recovery and containment)**.** *If $V(g) = 0$ at every stage and DualInterval evaluation returns finite upper enclosures $\overline{F} < \infty$ on each $\mathcal{I}_x$ (the finiteness half of (7), discharged in practice by the C^∞ operator set of Section 4.3), then the soft path recovers the hard-path conclusion of Theorem 1 exactly. The scores v_{mono}, v_{conv}, v_{irred} upper-bound the corresponding true pointwise violations on $\mathcal{I}_x$ (while v_{decay} and v_{leaf} are exact). Moreover, for every finite operator set $\mathcal{P}$ there exists $\varepsilon_{\mathrm{decay}} > 0$ small enough that every correction accepted by the hard certificates (a)–(b) has $V(g) = 0$ (and finite enclosures), so the soft search contains the hard one of Section 4.4.*

(See Appendix C.4 for the proof.) The soft scores therefore yield a *measured* admissibility gap rather than silent failure. Soft scores act as IA proxies for (A1)–(A4) (with positivity (A5) folded into the irreducible-floor gap and (A6) into finiteness of the DualInterval enclosure / C^2—in practice C^∞—operator set). Comparing soft-penalty search ($\lambda_a > 0$) against an unconstrained baseline ($\lambda_a = 0$), and optionally against a true hard reject filter, is one of the empirical questions the project settles.

5 Additional information

5.1 Collaborators welcome

Code skeleton: `src/modeling/` (modeling); `src/search/`, `src/systems/hpc/` (search)

The project splits into two fairly independent tracks, so a teammate could own either one. The *modeling* track ports and benchmarks transformer pretraining on Neocortex's Cerebras CS-3 systems (Section 3.1) and profiles it against our NVIDIA GPU runs (Section 3.2; useful background: PyTorch, or curiosity about the Cerebras software stack). The *search* track ports the genetic-programming symbolic-regression loop of Sections 4.3, 4.5 and 4.7 to AMA27 (Section 3.1) and profiles the DualInterval certification inner loop of Section 4.4 on AmpereOne against our AMD EPYC (x86) CPU baselines (useful background: C/C++ or Python performance work, or an interest in Arm/HPC profiling); a self-contained slice of it is benchmarking soft-penalty search against $\lambda_a = 0$ and against a hard reject filter (Section 4.4). No prior scaling-law expertise is needed for either; the main thing is an appetite for making code run fast on unusual hardware. The student skeleton paths for each track are listed in Appendix B.

5.2 Deliverables

Code skeleton: `src/systems/pipeline/run_discovery.py`

(1) An admissible boosted scaling law (Sections 4.2 and 4.6) fit on the collected training grid (Section 3.5); (2) a public release of the shape-constrained symbolic-regression library with the interval-arithmetic-certified admissibility checking of Section 4.4; and (3) a short paper on the method of Section 4. Notation, the description↔skeleton map, and the formal proofs appear in Appendices A to C.

6 References

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A Notation

Symbols used in Section 4 (and pointed to from the student stubs of Appendix B). Formal claims from Section 4 are proved in Appendix C.

Symbol	Meaning
N, D	Parameter count and training-token count (scale variables).
M	Tokens-per-parameter ratio $M = D/N$ (Section 3.4).
$\mathcal{H} = \{h_1, \dots, h_m\}$	Remaining hyperparameters; $m = \lvert\mathcal{H}\rvert$.
i	Hyperparameter index, $i = 1, \dots, m$ (distinct from data-point index ℓ).
H_h	Finite set of admissible values for coordinate $h \in \{N, D\} \cup \mathcal{H}$.
$\phi_h,\ \phi_i$	Strictly monotone preprocessing $\phi_h \colon H_h \to \mathbb{R}_{>0}$; $\phi_i \coloneqq \phi_{h_i}$.
$\mathbb{H}$	Discrete configuration space $\prod_h H_h$ (labeled grid).
$\widetilde{\mathbb{H}}$	Continuum domain of Sections 4.1 and 4.2: each scale ranges over $[x_{\min}, \infty)$, other coordinates over finite H_h; $\mathbb{H} \subset \widetilde{\mathbb{H}}$.
$\mathbf{h}$	Typical element of $\mathbb{H}$ or $\widetilde{\mathbb{H}}$ (coordinates N, D, h_i).
$L,\ \hat{L}$	True validation loss and its approximation.
$\mathcal{D},\ n,\ \ell$	Labeled dataset $\{(\mathbf{h}_\ell, L(\mathbf{h}_\ell))\}_{\ell=1}^{n}$; $n = \lvert\mathcal{D}\rvert$; data-point index ℓ.
x	Generic scale variable, $x \in \{N, D\}$.
t	Dummy limit variable along a scale axis (e.g. $\hat{L}(\mathbf{h})\vert_{x=t}$).
$x_{\min},\ x_{\max}$	$\min H_x$ and $\max H_x$.
$\mathcal{S}$	Set of admissible scaling laws (Section 4.2).
$L_\infty,\ L_\infty^{(j)}$	Joint irreducible-loss floor; joint floor of $\hat{L}^{(j)}$ ($L_\infty^{(j)} = E$ under the hard path).
L_x^∞	Marginal floor along axis x (1) (depends on the frozen coordinates).
$c_x,\ c_x^{(0)}$	Asymptotic power-law exponent along x (A4); stage-0 value $c_N^{(0)} = -\alpha$, $c_D^{(0)} = -\beta$.
$j,\ K$	Boosting stage index; number of correction stages after stage 0 (Section 4.5, Algorithm 1).
$\hat{L}_j,\ \hat{L}^{(j)}$	Stage-j correction (subscript) and cumulative ensemble through stage j (superscript); $\hat{L}^{(0)} = \hat{L}_0$.
$A, B, C_i, E, \alpha, \beta, \gamma_i$	Positive coefficients/exponents of the stage-0 Chinchilla law (3); joint floor $L_\infty = E$.
δ_j	Huber clipping threshold at stage j (4).
$\tilde{r}_j^{(\ell)}$	Huber pseudo-residual for data point ℓ at stage j (5).
$\mathcal{P},\ p$	Finite set of admissible powers $\mathcal{P} \subset \mathbb{R} \setminus \{0\}$; generic element p.
pow_p	Power primitive $\mathrm{pow}_p(z) = \lvert z\rvert^p$ for $z > 0$.
$g,\ F$	Candidate expression tree; ensemble-plus-candidate $F = \hat{L}^{(j-1)} + g$.
$c_{x,j},\ \mathrm{ord}(g, x)$	Asymptotic exponent of correction/candidate g along x; $c_{x,j} = \mathrm{ord}(g, x)$ in (6)(v).
$\mathcal{I}_x$	Compact certification box $[x_{\min}, x_{\max}]$ along scale x (written I_x in the skeleton); with other coordinates over finite H_h, the product of boxes is a compact continuum slice of $\widetilde{\mathbb{H}}$ covering the observed grid (Section 4.4); the tail $x > x_{\max}$ is controlled by ord.

Symbol	Meaning
$\underline{f},\ \overline{f}$	Interval-arithmetic lower/upper enclosures of f on $\mathcal{I}_x$ (global over that box for fixed x; soft scores take $\max_{x\in\{N,D\}}$ of the per-axis gaps). $\underline{F} > L_\infty$ certifies (A3)/(A5) on the box; $\overline{F} < \infty$ with the smooth operator set certifies (A6).
$a,\ v_a(g),\ V(g)$	Soft-path axiom index $a \in \{\text{mono}, \text{conv}, \text{irred}, \text{decay}, \text{leaf}\}$; per-axiom violation score; aggregate $V = \sum_a v_a$ (9).
$\varepsilon_{\text{decay}}$	Positive margin in v_{decay} so zero score implies strict power-law decay (9).
$\text{leaves}(g)$	Set of variable leaves of expression tree g.
λ_a	Positive penalty weight for axiom a.
$F_j(g)$	Soft-path penalized fitness at stage j (10) (distinct from F).

B Code skeleton map

The repository ships a student implementation skeleton under `src/`, grouped into five packages (`src/README.md`). Blue links open the corresponding paths on GitHub (`main`). Each technical subsection in the main text also points to those files via the *Code skeleton* lines; symbols used there are listed in Appendix A. The stage-0 baseline used by the boosting stubs is Proposition 1; proofs of all formal claims are in Appendix C.

Description	Package	Primary paths
Section 4.1	`src/scaling/`	`src/scaling/setup/configuration.py`, `src/scaling/setup/constants.py`
Section 3.5	`src/scaling/`	`src/scaling/data/scaling_dataset.py`, `src/scaling/data/corpora.py`
Section 4.2	`src/constraints/`	`src/constraints/axioms/admissibility.py`
Section 4.3	`src/search/`	`src/search/baselines/`, `src/search/residuals/`, `src/search/expression/`
Section 4.4	`src/constraints/`, `src/search/`	`src/constraints/axioms/stage_conditions.py`, `src/constraints/certificates/`, `src/search/hard_path/`
Section 4.5, Alg. 1	`src/search/`	`src/search/boosting/`
Section 4.6	`src/constraints/`	`src/constraints/guarantee/invariants.py`
Section 4.7	`src/search/`	`src/search/soft_path/`
Section 3.3 (optional JAX)	`src/constraints/`	`src/constraints/certificates/jax_backend.py`
Section 3.4	`src/modeling/`	`src/modeling/models/config.py`, `src/modeling/models/transformer.py`
Section 3.1 (Neocortex)	`src/modeling/`	`src/modeling/training/`
Sections 3.1 and 3.2 (AMA27/CPU)	`src/systems/`	`src/systems/hpc/`
Section 5.2	`src/systems/`	`src/systems/pipeline/run_discovery.py`

C Proofs

Proofs follow the dependency order of Section 4: stagewise equivalence, then the certificate lemmas used by the hard filter, then the hard-path theorem, then the soft-path proposition. Stage-0 admissibility (Proposition 1) is independent of the boosting loop and is proved last.

C.1 Stagewise equivalence

Proof of Proposition 2. Write $G := \hat{L}^{(j-1)}$ and $g := \hat{L}_j$, so $\hat{L}^{(j)} = G + g$. Throughout, derivatives along $x \in \{N, D\}$ hold the other coordinates fixed.

(A1)–(A2). $\partial(G+g)/\partial x = \partial G/\partial x + \partial g/\partial x$ and likewise for second derivatives. Since $G \in \mathcal{S}$, one has $\partial G/\partial x \le 0$ and $\partial^2 G/\partial x^2 \ge 0$ on $[x_{\min}, \infty)$. Thus $\partial(G+g)/\partial x \le 0$ on that half-line if and only if (i) holds, and $\partial^2(G+g)/\partial x^2 \ge 0$ if and only if (ii) holds.

(A5) and joint floor (A3). Positivity $G + g > 0$ on $\widetilde{\mathbb{H}}$ is exactly (iii). The joint-floor inequality $G + g > L_\infty$ on $\widetilde{\mathbb{H}}$ is exactly (iv). (Here L_∞ is the joint floor already certified for G; (iv) keeps it for the update. When $L_\infty > 0$, (iv) implies (iii).)

(A6). If G is C^2 on $[x_{\min}, \infty)$ and g is C^∞ on $[x_{\min}, x_{\max}]$ with finite boundary values (vi), then $G+g$ is C^2 on the certification box. On the half-line beyond $x_{\max}$, the same C^2 regularity follows once g is realized by a finite expression tree from $\{+, -, \times, \div\} \cup \{\mathrm{pow}_p\}$ (each primitive is C^∞ on $(0, \infty)$), which is the setting of Section 4.3. Finiteness of $(G+g)(x_{\min}, \cdot)$ is the restriction of (vi) together with $G \in \mathcal{S}$; the bound $(G+g)(x_{\min}, \cdot) > L_\infty$ is the restriction of (iv).

(A3) marginal floors and (A4). Because $G \in \mathcal{S}$, the marginal floor $L_x^\infty(G)$ exists and the excess $G|_{x=t} - L_x^\infty(G)$ is asymptotic to a constant times $t^{c_x^{(0)}}$ with $c_x^{(0)} < 0$. Condition (v) gives $g|_{x=t} = O(t^{c_{x,j}})$ with $c_{x,j} < c_x^{(0)}$, so $g|_{x=t} = o\big(t^{c_x^{(0)}}\big)$. Therefore

$$\lim_{t\to\infty} (G+g)\big|_{x=t} \;=\; L_x^\infty(G),$$

the marginal floor is unchanged, and

$$\lim_{t\to\infty} \frac{\log\big((G+g)|_{x=t} - L_x^\infty(G)\big)}{\log t} \;=\; c_x^{(0)}.$$

Conversely, if $G + g \in \mathcal{S}$ with the same joint floor and the same asymptotic rate $c_x^{(0)}$, the difference $g = (G+g) - G$ must be $o(t^{c_x^{(0)}})$ along each axis (else the leading excess exponent of $G+g$ would differ from that of G), which is (v); the remaining items (i)–(iv) and (vi) are the rearrangements already used above.

Thus $G + g \in \mathcal{S}$ with joint floor L_∞ if and only if (6) holds. □

C.2 Interval and structural certificates

Lemma 1 (Soundness of the interval tests). *Suppose forward-mode DualInterval evaluation returns sound enclosures $\underline{f} \le f \le \overline{f}$ on $\mathcal{I}_x$ for $f \in \{F, \partial F/\partial x, \partial^2 F/\partial x^2\}$. If (7) holds, then on $\mathcal{I}_x$: $\partial F/\partial x \le 0$, $\partial^2 F/\partial x^2 \ge 0$, and $F > L_\infty$, and $|F| < \infty$. In particular, writing $F = G + g$ with $G = \hat{L}^{(j-1)}$,* *conditions (i)–(iv) and the finiteness half of (vi) hold on the continuum configurations covered by the product of boxes $\mathcal{I}_x$ (a compact slice of $\widetilde{\mathbb{H}}$). When $L_\infty > 0$, the floor test also yields positivity (A5) / (iii). Smoothness (A6) on the box follows from the C^∞ operator set once finiteness holds.*

Proof. Soundness means every true value on $\mathcal{I}_x$ lies in the returned interval. Hence $\overline{\partial F/\partial x} \leq 0$ forces $\partial F/\partial x \leq 0$ everywhere on $\mathcal{I}_x$; $\underline{\partial^2 F/\partial x^2} \geq 0$ forces $\partial^2 F/\partial x^2 \geq 0$; and $\underline{F} > L_\infty$ forces $F > L_\infty$ on the box (hence also $F > 0$ when $L_\infty > 0$), which is (iv) and yields (iii). Boundedness of $\overline{F}$ is the finiteness claim of (vi). The argument is one-sided: failure of (7) does not refute (6). □

Lemma 2 (Structural exponent)**.** *Let g be a finite expression tree over $\{+, -, \times, \div\} \cup \{\mathrm{pow}_p : p \in \mathcal{P}\}$ in which the scale leaf x appears. Define* $\mathrm{ord}(g, x)$ *recursively as in Section 4.4(b). Then* $g(\mathbf{h})|_{x=t} = \Theta\big(t^{\mathrm{ord}(g,x)}\big)$ *as $t \to \infty$ (leading homogeneous term), and hence* (6)*(v) holds with* $c_{x,j} = \mathrm{ord}(g, x)$ *if and only if* $\mathrm{ord}(g, x) < c_x^{(0)}$.

Proof. Proceed by induction on tree size. Constants and leaves other than x are $O(1) = \Theta(t^0)$. A factor $\mathrm{pow}_p(x)$ contributes $\Theta(t^p)$. Products multiply leading powers (exponents add); quotients subtract; a sum is dominated by the child of largest exponent, so the inductive clause "sums take the maximum" records the leading order. (If two children share the same maximal exponent their leading coefficients might cancel, in which case the true order is strictly smaller; the recursive ord is then a sound upper bound, and the test $\mathrm{ord}(g, x) < c_x^{(0)}$ remains sufficient for (v).) Thus $g|_{x=t} = O\big(t^{\mathrm{ord}(g,x)}\big)$, and whenever there is no exact leading-term cancellation one has matching Θ asymptotics. Condition (v) asks for some $c_{x,j} < c_x^{(0)}$ with $g = O(t^{c_{x,j}})$, which holds with $c_{x,j} = \mathrm{ord}(g, x)$ precisely when $\mathrm{ord}(g, x) < c_x^{(0)}$. □

C.3 Hard-path boosting guarantee

Proof of Theorem 1. Induct on the stage index j. For $j = 0$, the claim is Proposition 1: $\hat{L}_0 \in \mathcal{S}$ with $L_\infty^{(0)} = E$ and the stated $c_x^{(0)}$.

Fix $j \geq 1$ and assume the claim holds through stage $j - 1$. Certificates (a)–(b) accept $g = \hat{L}_j$, so by Lemma 1 conditions (i)–(iv) and (vi) hold on each certification box $\mathcal{I}_x$ (hence (A1)–(A3), (A5)–(A6) on that compact slice of $\widetilde{\mathbb{H}}$), and by Lemma 2 condition (v) holds on the half-line. The box covers the observed grid; on $x > x_{\max}$, (v) together with $G = \hat{L}^{(j-1)} \in \mathcal{S}$ implies that the correction is negligible relative to the leading excess of G, so (A1)–(A2) of G persist for large x (the derivatives of an $O(t^c)$ term with $c < c_x^{(0)} < 0$ are o of those of the leading $t^{c_x^{(0)}}$ excess), and C^2 regularity (A6) extends by the operator set. Proposition 2 therefore yields $\hat{L}^{(j)} \in \mathcal{S}$ with the same joint floor $L_\infty^{(j)} = L_\infty^{(j-1)} = E$. The marginal-floor / rate calculation in the proof of Proposition 2 gives (8). □

C.4 Soft-path guarantees

Proof of Proposition 3. By definition each $v_a(g) \geq 0$, and $V(g) = \sum_a v_a(g)$. Hence $V(g) = 0$ if and only if every score vanishes.

Recovery. $v_{\mathrm{mono}}(g) = 0$ iff $\overline{\partial F/\partial x} \leq 0$ for all $x \in \{N, D\}$, which is the monotonicity half of (7); likewise $v_{\mathrm{conv}} = 0$ recovers the convexity test and $v_{\mathrm{irred}} = 0$ recovers the floor test of Lemma 1 (hence also positivity (A5) when $L_\infty > 0$). The remaining interval test $\overline{F} < \infty$ is not encoded as a soft score v_a: under the smooth operator set of Section 4.3, a completed DualInterval evaluation already yields finite enclosures, so (A6) / (vi) holds whenever the IA layer succeeds. $v_{\mathrm{leaf}}(g) = 0$ iff both scale variables appear as leaves. $v_{\mathrm{decay}}(g) = 0$ iff $\mathrm{ord}(g, x) \leq c_x^{(0)} - \varepsilon_{\mathrm{decay}} < c_x^{(0)}$, which is a strict strengthening of (v).

Thus $V(g) = 0$ together with finite DualInterval enclosures implies certificates (a)–(b) (with the decay margin), and Theorem 1 applies stage by stage.

Measured gaps. Soundness of DualInterval enclosures yields $\partial F/\partial x \le \overline{\partial F/\partial x}$ and $\partial^2 F/\partial x^2 \ge \underline{\partial^2 F/\partial x^2}$ on $\mathcal{I}_x$, so

$$\max\bigl(0, \partial F/\partial x\bigr) \;\le\; \max\bigl(0, \overline{\partial F/\partial x}\bigr)$$

pointwise on the box (and likewise for the convexity and floor / positivity gaps). Taking maxima over $x \in \{N, D\}$ shows that v_{mono}, v_{conv}, and v_{irred} upper-bound the corresponding true violations on the compact slice of $\widetilde{\mathbb{H}}$. The quantities $\operatorname{ord}(g, x)$ and $\operatorname{leaves}(g)$ are exact combinatorial reads of the tree, so v_{decay} and v_{leaf} are exact.

Containment. The attainable values of $\operatorname{ord}(g, x)$ for trees over a finite power set $\mathcal{P}$ form a finite subset of $\mathbb{R}$. Let Δ be the minimal positive gap between $c_x^{(0)}$ and any attainable exponent strictly below it (or $\Delta = 1$ if no such exponent exists). Choosing $0 < \varepsilon_{\text{decay}} < \Delta$ forces: every g with $\operatorname{ord}(g, x) < c_x^{(0)}$ automatically satisfies $\operatorname{ord}(g, x) \le c_x^{(0)} - \varepsilon_{\text{decay}}$. Therefore every hard-accepted correction has $v_{\text{decay}} = 0$ as well as vanishing mono/conv/irred/leaf scores, i.e. $V(g) = 0$, and already satisfies $\overline{F} < \infty$ by (7), so it is a zero-penalty feasible point of (10). □

C.5 Stage-0 Chinchilla admissibility

Proof of Proposition 1. Write the hyperparameter remainder

$$R(\mathbf{h}) \;\coloneqq\; \sum_{i=1}^{m} \frac{C_i}{\phi_i(h_i)^{\gamma_i}} + E,$$

so $\hat{L}_0 = AN^{-\alpha} + BD^{-\beta} + R(\mathbf{h})$. Every summand of R is strictly positive on $\widetilde{\mathbb{H}}$, hence $R(\mathbf{h}) > E > 0$. Shape axioms (A1), (A2), (A4), and (A6) are imposed only along $x \in \{N, D\}$ with the other coordinates held fixed; under that restriction R is constant in x.

(A1)–(A2). Fix $x = N$ and freeze $(D, h_1, \ldots, h_m)$. Then

$$\frac{\partial \hat{L}_0}{\partial N} \;=\; -\alpha A\, N^{-\alpha-1} \;<\; 0, \qquad \frac{\partial^2 \hat{L}_0}{\partial N^2} \;=\; \alpha(\alpha+1)\, A\, N^{-\alpha-2} \;>\; 0$$

for all $N \ge N_{\min} > 0$. The same identities with (A, α, N) replaced by (B, β, D) give (A1)–(A2) along D.

(A3). Set $L_\infty \coloneqq E > 0$. For every continuum configuration with $N, D \ge$ their minima (and hyperparameters in their H_h),

$$\hat{L}_0(\mathbf{h}) - E \;=\; \frac{A}{N^\alpha} + \frac{B}{D^\beta} + \sum_{i=1}^{m} \frac{C_i}{\phi_i(h_i)^{\gamma_i}} \;>\; 0,$$

so the joint-floor inequality of (A3) holds strictly on $\widetilde{\mathbb{H}}$. Along N with the other coordinates fixed,

$$L_N^\infty \;=\; \lim_{t\to\infty} \hat{L}_0\big|_{N=t} \;=\; \frac{B}{D^\beta} + R(\mathbf{h}) \;=\; \frac{B}{D^\beta} + \sum_{i=1}^{m} \frac{C_i}{\phi_i(h_i)^{\gamma_i}} + E,$$

which is finite and satisfies $L_N^\infty \ge E = L_\infty$ (and is strictly larger whenever $B > 0$ or $m \ge 1$). The marginal floor L_D^∞ is analogous.

(A4). With the same frozen coordinates, $\hat{L}_0\big|_{N=t} - L_N^{\infty} = A\,t^{-\alpha}$, hence

$$\lim_{t\to\infty} \frac{\log\big(\hat{L}_0\big|_{N=t} - L_N^{\infty}\big)}{\log t} \;=\; -\alpha \;=\; c_N^{(0)} \;<\; 0.$$

The D-axis calculation yields $c_D^{(0)} = -\beta < 0$.

(A5)–(A6). Positivity (A5) is immediate from $\hat{L}_0(\mathbf{h}) > E > 0$ on $\widetilde{\mathbb{H}}$. For (A6): on $[N_{\min}, \infty)$ the map $N \mapsto AN^{-\alpha}$ is C^{∞} (hence C^2), the remaining terms are independent of N, and $\hat{L}_0(N_{\min}, \cdot) = AN_{\min}^{-\alpha} + BD^{-\beta} + R(\mathbf{h})$ is finite (and strictly larger than E by (A3)). The same holds with N replaced by D.

Thus $\hat{L}_0$ satisfies (A1)–(A6), i.e. $\hat{L}_0 \in \mathcal{S}$. $\square$

Remark. If the hyperparameter block in (3) is replaced by nonnegative quadratic penalties in $\log \phi_{\mathrm{lr}}(\mathrm{lr})$ and $\log \phi_{\mathrm{wd}}(\mathrm{wd})$ (as in the 4D U-shaped baseline of Section 4.3), those terms remain constant along N and D. Whenever $A, B, E, \alpha, \beta > 0$, the same derivative and asymptotic calculations apply verbatim, so (A1)–(A6) continue to hold with the same $L_\infty = E$ and $c_x^{(0)}$.